NFCAP LMS — Build Plan

A web application for apprentices, NFCAP admins and WIOA funding staff.

1. At a Glance

Scope	Effort	Timeline	Hard gate
65 features in **9 phases**	**1,431 hours** (1,278 core + 12% buffer)	**~18 weeks / ~4 months** with 2 engineers *(team size: [CONFIRM])*	Real WIOA data cannot enter the system until the **Workforce Board Data Agreement** is signed (Phase 8)

This is a **web application only**. A mobile app is deliberately not part of the build; it is a future add-on, only if needed. Eight further items (**add-ons**) are held outside this scope and quoted separately — see section 7.

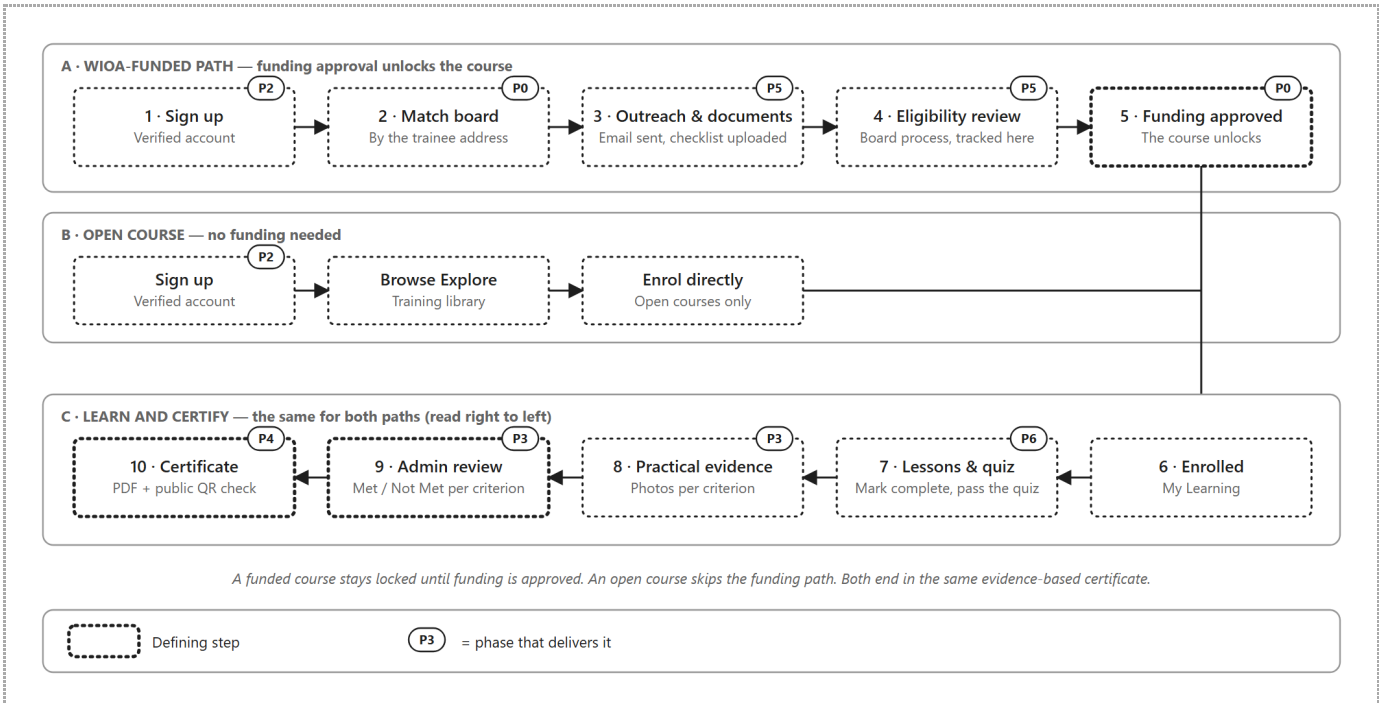
Two phases define the product: Phase 5, which turns WIOA funding into a clear, working process, and Phase 3, which makes certification depend on real, assessed evidence. Funding gets a trainee into a course; evidence is what earns the certificate.

How to read this plan. Sections 2 to 4 show what the system does: the trainee's journey, how staff and admins run it, and the 13 modules it is made of. Section 6 shows how and when it is built, phase by phase. Every journey step and module names the phases that build it.

2. The Product Journey

NFCAP LMS takes a trainee from funding, to a course, to a certificate. A **WIOA-funded** trainee goes through steps 1 – 5, and funding approval unlocks the course. A trainee on an **open course** signs up and enrols directly at step 6. After that, everyone follows the same steps 6 – 10.

NFCAP does not run the WIOA process; the workforce board does. There is no link to any government system. The connection is human: an email, a phone call and status updates entered by NFCAP staff.



The ten steps

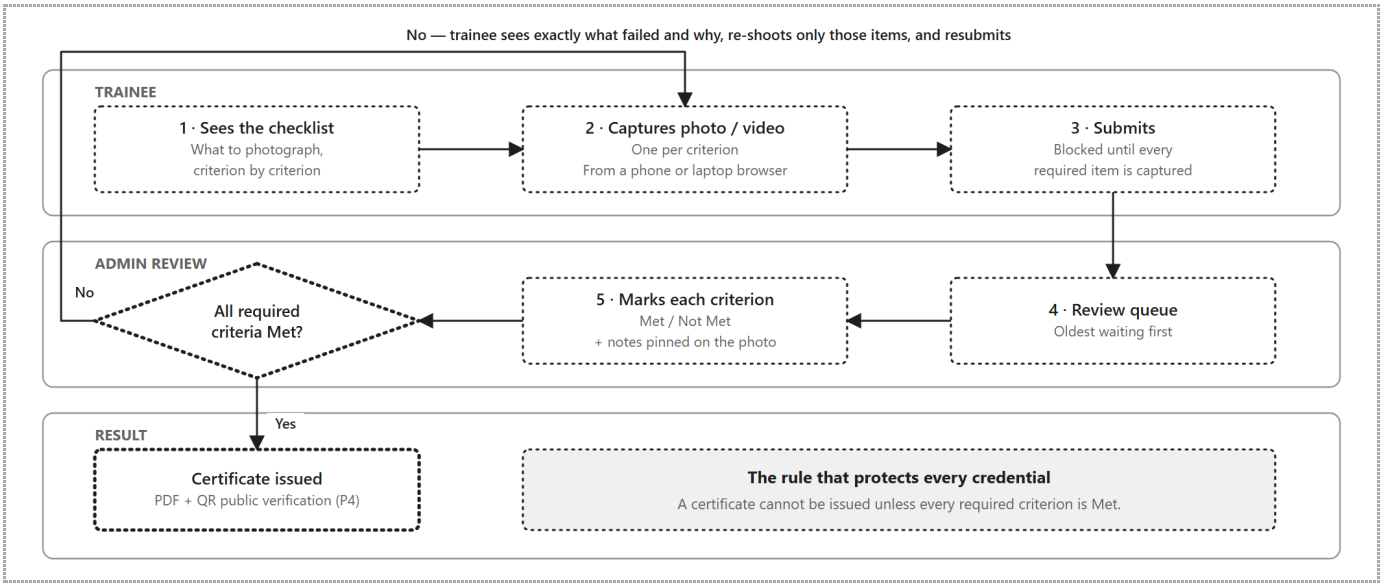
#	Step	What happens	Who acts	Built in
1	**Sign up**	A short profile only: name, date of birth, email and phone, address, employment and unemployment-insurance status, training interest. The address drives step 2. The full workforce-board application is **not** part of sign-up.	Trainee	Phase 2
2	**Match board**	The address picks the workforce board, career centre and contact. If there is no match, staff resolve it by hand.	Automatic	Phase 0
3	**Outreach and documents**	The trainee reviews and sends a pre-written email to the board (logged). They upload files against **that board's checklist** (ID, proof of residency, right to work, resume, unemployment letter, Selective Service, veteran status). Staff accept, reject or ask for a resubmission.	Trainee, Partner Staff	Phase 5
4	**Eligibility review**	Orientation, CalJOBS account, the **application form** (demographics, family income, education, public assistance, barriers, signature), intake appointment and the eligibility decision all happen **with the board, outside the system**. Staff record each status here.	Board; staff record	Phase 5
5	**Funding approved**	Training programme chosen, funding requested, case-manager review, training authorisation and tuition approval, each tracked as its own milestone. Approval unlocks that one course for that trainee.	Board; Coordinator records	Phase 5
6	**Enrolled**	The course appears in My Learning.	Trainee	Core system
7	**Lessons and quiz**	Lessons marked complete, then a knowledge quiz.	Trainee	Phase 6
8	**Practical evidence**	Photo or video against each named criterion; no submission until every required item is captured.	Trainee	Phase 3
9	**Admin review**	Each criterion marked Met or Not Met, with notes pinned on the photo. Failures send the trainee back to re-shoot only those items.	Admin	Phase 3
10	**Certificate**	Issued only if every required criterion is Met. A PDF with a public QR code anyone can verify.	Automatic	Phase 3, Phase 4

Only one workforce board's packet (Tri-Valley Career Center, Alameda County) is on file so far. Other boards will have their own checklists; see the decisions in section 8.

The rule that matters most: a funded course stays locked until tuition is approved, so no one starts training that funding might later refuse to pay for. Open courses never touch the funding process.

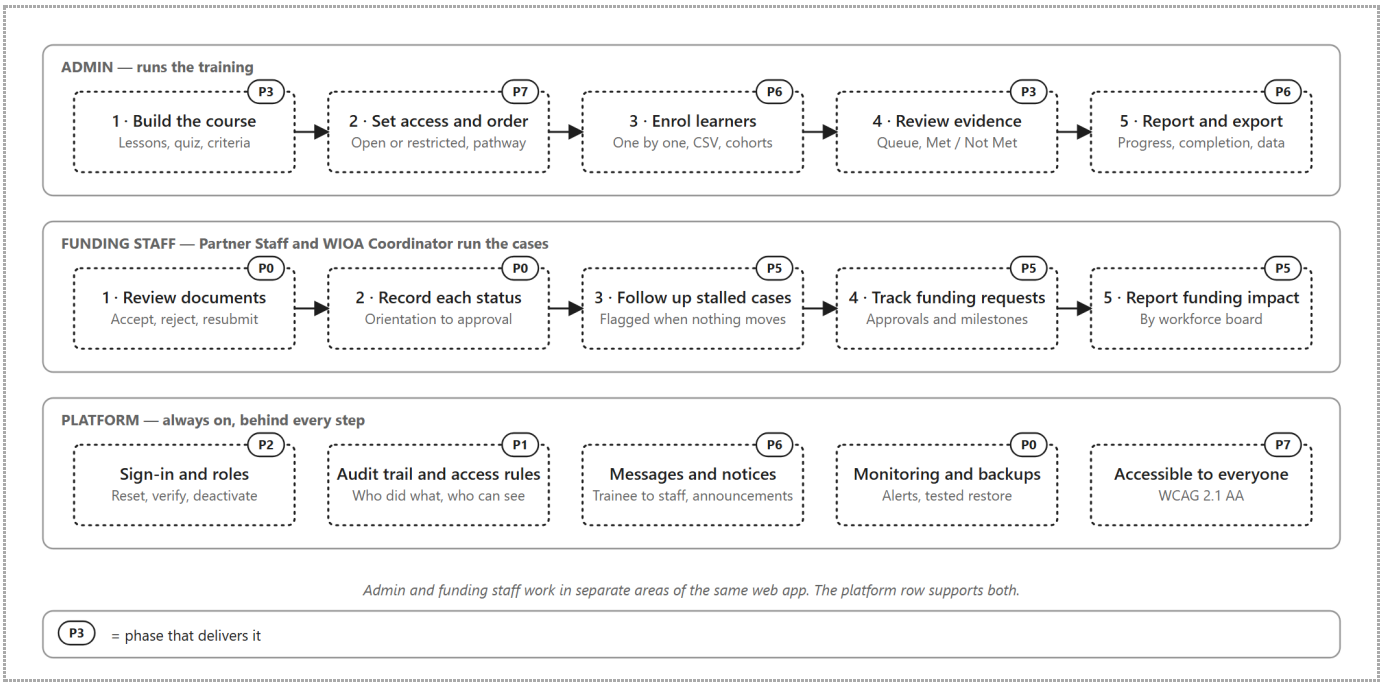
Steps 8 – 10 in detail

Phase 3 builds this loop: structured, photographic proof of real work, assessed criterion by criterion, with certification that cannot be issued unless the assessment is complete.



3. How Staff and Admins Run It

The journey above is the trainee's view. Behind it, NFCAP staff work in separate areas of the same web app: Admin runs the training, and funding staff run the WIOA cases. A platform layer supports both.



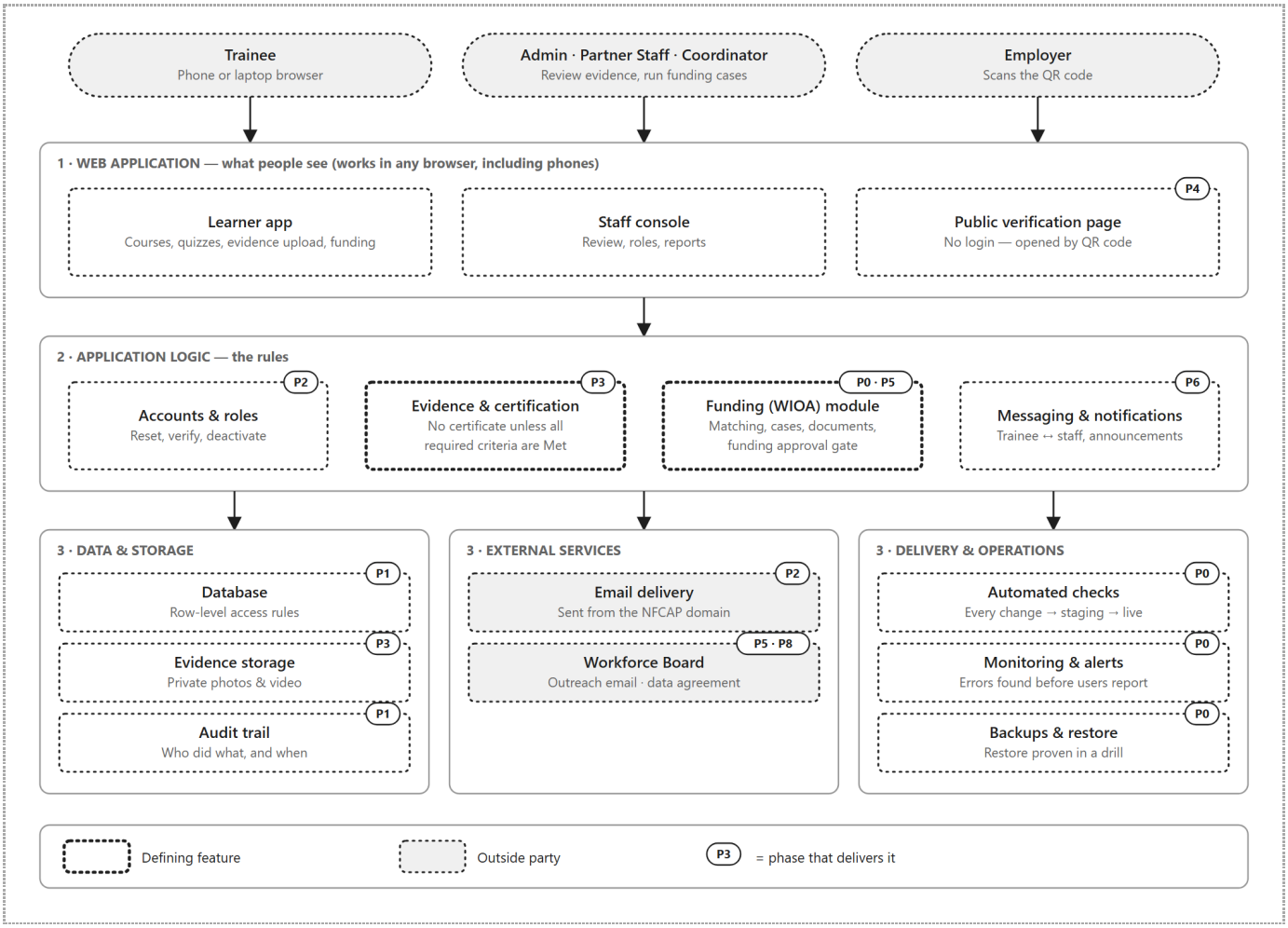
Role	Sees and does
Trainee	Own stage, next step, documents, appointments, lessons, evidence
Partner Staff	Only their assigned trainees: progress, missing documents, follow-up notes. No funding amounts
WIOA Coordinator	All funding cases, communications, appointments and funding requests
Admin	Courses and pathway, learners and cohorts, evidence review, roles, reports, plus workforce boards, contacts and document checklists

4. Product Modules

Everything in the plan belongs to one of 13 modules, so nothing is left out. Each row shows what the module does and which phases and hours deliver it.

Module	Used by	What it does	Phases	Hours
Accounts & Profile Learner	Trainee	Sign in and manage an account: password reset, email verification, account settings, secure sessions	Phase 2	56
Learning & Quizzes Learner	Trainee	Structured lessons marked complete by the learner, then a knowledge quiz with several question types, question banks, random order and attempt limits	Phase 6	32
Evidence & Portfolio Learner	Trainee	Trainees submit real photo and video evidence against each named criterion, get clear feedback, resubmit only what failed, and keep a portfolio of verified work (optionally shareable)	Phase 3, Phase 4	126
Certificates Learner	Trainee, employers	A downloadable PDF certificate with a public QR verification page, and optional expiry and renewal reminders	Phase 4	62
WIOA Funding Funding	Trainee, Partner Staff, Coordinator	Match a trainee to a workforce board, send the outreach email, upload documents against the board checklist, and track every status through to funding approval, with stalled-case alerts and impact reporting	Phase 0, Phase 5, Phase 8	166
Curriculum & Pathway Admin	Admin	Build courses (sections, lessons, quiz, assessment criteria) and order the pathway, including reordering by keyboard	Phase 1, Phase 3, Phase 7	86
Evidence Review Admin	Admin	A review queue where each criterion is marked Met or Not Met with notes pinned on the photo; a certificate is issued only when every required criterion is Met	Phase 3	98
Users & Cohorts Admin	Admin	Assign roles, offboard users safely, enrol groups by CSV, and organise trainees into cohorts with deadlines	Phase 2, Phase 6	78
Reporting & Data Admin	Admin	Completion trends, time to complete, at-risk flags and data export	Phase 1, Phase 6	50
Communication Shared	Everyone	Messaging between trainee and staff, announcements, notifications, and reliable email from the NFCAP domain	Phase 2, Phase 6	84
Security & Compliance Platform	Everyone	Database-level access rules, a full audit trail, no Social Security numbers stored, a retention policy, terms and privacy policy, and an independent security review	Phase 1, Phase 8	156
Quality & Operations Platform	NFCAP and the build team	Automated tests and checks on every change, monitoring, staging, tested backups, load testing, a runbook and acceptance testing before go-live	Phase 0, Phase 8	202
Accessibility Platform	Everyone	WCAG 2.1 AA across every screen, tested with keyboard and screen reader	Phase 7	82
		65 features, 13 modules		**1,278**

5. System Architecture



Dotted boxes are parts of the system, tagged with the phase that delivers them (P0–P8). Shaded boxes are outside parties. Bold outlines mark the defining features.

6. Build Plan

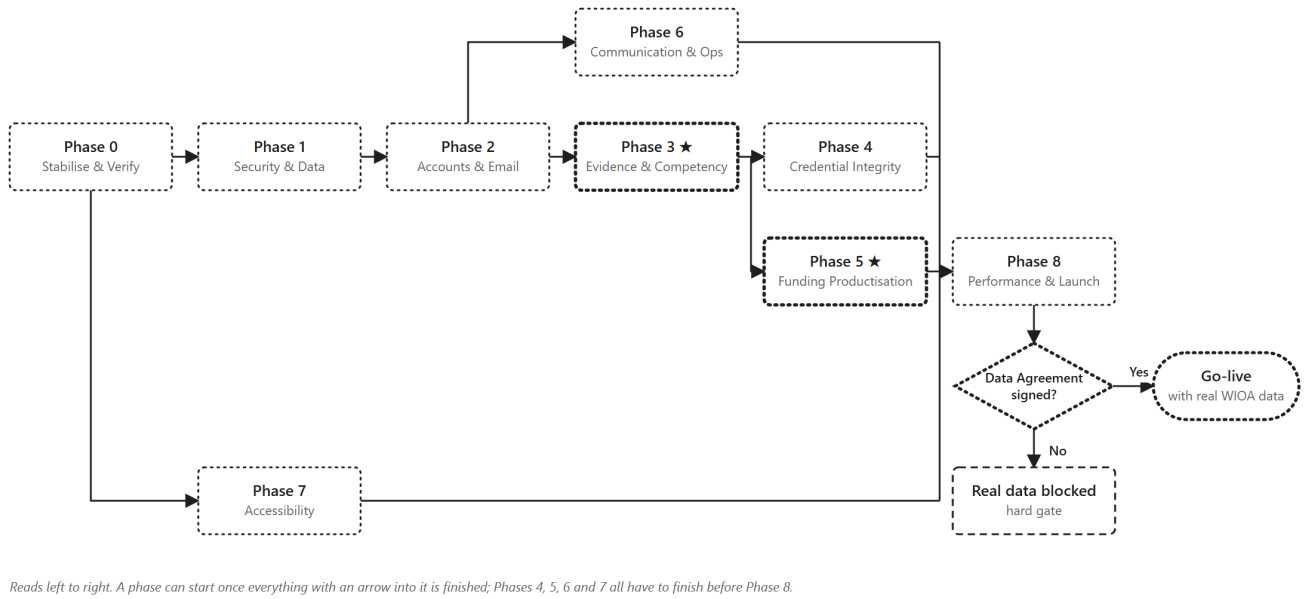
Nine phases in four stages, built back to back by a two-engineer team (80 team-hours a week). Each phase names the journey steps it delivers.

Phase	What it delivers	Journey	Features	Hours	~Weeks
	Stage 1 · Foundation — tested, secure and operable by real people		**22**	**406**	**5.1**
0	Quality Foundations & Delivery Pipeline Automated tests, checks on every change, monitoring, staging and proven backups; sign off the WIOA funding module.	Steps 2–5 verified	6	142	1.8
1	Security & Data Protection Put protections in place before any real trainee or WIOA data enters the system.	All steps protected	9	156	1.9
2	Accounts, Access & Email The everyday account functions every real service needs: email, password reset, roles, offboarding.	Step 1	7	108	1.4
	Stage 2 · Core Product — real evidence, gated certification, verifiable credentials, productised funding		**23**	**470**	**5.9**

Phase	What it delivers	Journey	Features	Hours	~Weeks
3	Evidence & Competency ★ Real photo evidence, assessed criterion by criterion. Certification only when every required criterion is Met.	Steps 8–10	12	260	3.3
4	Credential Integrity & Portability A credential trainees can hold, prove and carry: a PDF certificate with public QR verification.	Step 10	4	80	1.0
5	Funding Module Productisation ★ Make WIOA funding a clear, working process for trainees and staff: status, delivered email, impact reporting.	Steps 3–5	7	130	1.6
	Stage 3 · Operations & Access — messaging, cohorts, reporting; accessible to everyone		**14**	**286**	**3.6**
6	Communication & Operations Day-to-day operations: messaging, announcements, cohorts, reporting and a quiz that cannot be gamed.	Step 7, messaging	8	184	2.3
7	Accessibility (WCAG 2.1 AA) Bring the whole interface to WCAG 2.1 AA, very likely required for WIOA-funded providers.	All steps	6	102	1.3
	Stage 4 · Launch — load-tested, independently reviewed, signed off, live		**6**	**116**	**1.4**
8	Performance, Hardening & Launch Prove it performs and is secure at scale, sign the data agreement, and go live.	All steps (launch)	6	116	1.4
	Core total		**65**	**1,278**	**16.0**
	+ 12% risk buffer			153	1.9
	Project total			**1,431**	**17.9**

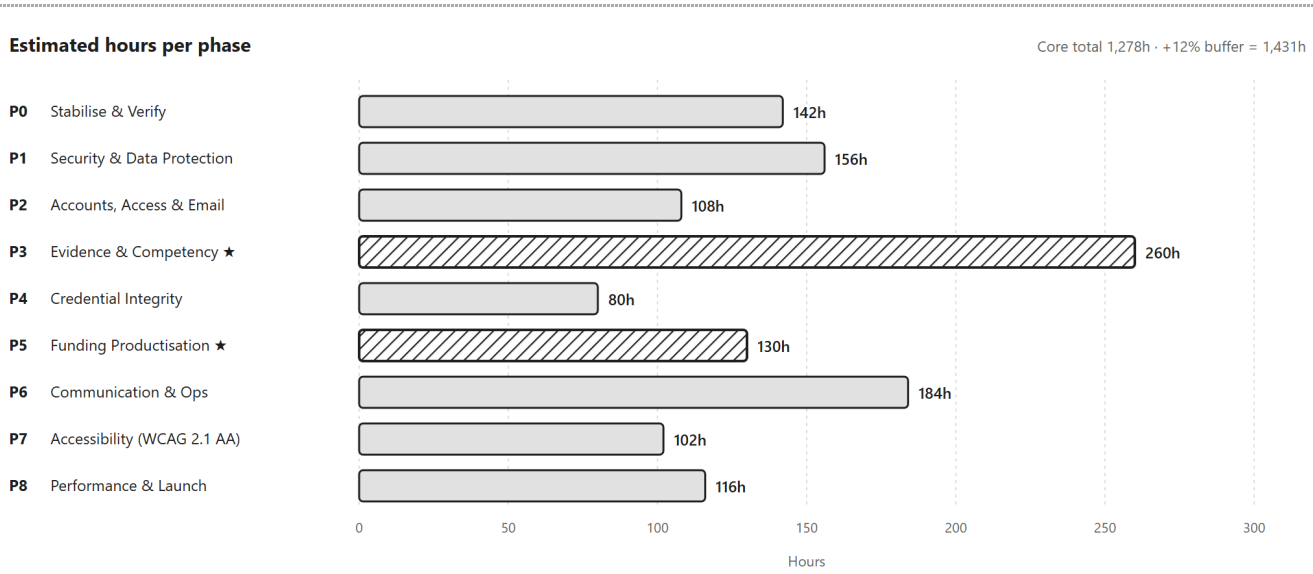
In person-days: 159.8 core + 19.2 buffer = **178.9**. *Cost figures (USD) and the blended hourly rate are intentionally omitted from this client copy; see the commercial proposal for pricing.*

Order and dependencies. Phases are numbered in build order, but not every phase depends on the one before it. Phases 4 and 5 both build on Phase 3, and Phase 7 only needs the Phase 0 automated checks. Everything converges on Phase 8. Real trainee data, especially WIOA funding data, cannot safely or legally enter the system until Phase 1 (security) is in place and the Phase 8 data agreement is signed, so every earlier phase is built and tested without real data.



Reads left to right. A phase can start once everything with an arrow into it is finished; Phases 4, 5, 6 and 7 all have to finish before Phase 8.

Effort by phase.



The plan for each phase follows. Full descriptions, roles, complexity and risk notes for every feature are in the Feature Detail appendix.

Phase 0 — Quality Foundations & Delivery Pipeline

Foundation · 6 features · **142h** · ~1.8 weeks · **Journey:** Steps 2–5 verified

Goal: Automated tests, checks on every change, monitoring, staging and proven backups; sign off the WIOA funding module.

Module	Feature	Hours
Verification	WIOA Funding Module Sign-Off	24
QA	Automated Test Suite — Critical Paths	40
DevOps	Automated Checks on Every Change (CI)	20
	Staging Environment & Migration Strategy	20

Module	Feature	Hours
Reliability	Error Handling, Logging & Monitoring	24
	Backup, Restore & Recovery Drill	14

Done when

- ☐ The WIOA funding module is verified end to end and signed off
- ☐ Automated tests cover enrollment gating, evidence → certificate, and funding status changes
- ☐ A broken change cannot reach the live site
- ☐ Errors are tracked and alerted on; a restore from backup has been proven in a drill

Phase 1 — Security & Data Protection

Foundation · 9 features · **156h** · ~1.9 weeks · **Journey:** All steps protected

Goal: Put protections in place before any real trainee or WIOA data enters the system.

Module	Feature	Hours
Security	Database Access-Rule Approach — Decision	16
	Row-Level Access Rules (RLS)	36
	Secrets, Environment & Rate Limiting	18
Audit	Platform-Wide Audit Trail	24
Privacy	No-SSN Rule & PII Boundary	12
	Data Retention & Destruction Policy	20
Legal	Terms of Service & Privacy Policy	10
Architecture	Pathway Structure Seam	12
	Hour-Derivable Progress History	8

Done when

- ☐ Database-level access rules are in place, or a documented alternative with compensating controls
- ☐ Sensitive activity is fully audit-logged and exportable
- ☐ The system cannot store an SSN; retention and destruction run automatically
- ☐ Terms and privacy policy are live; secrets and rate limits are hardened

> ⚠️ **Decision needed:** does NFCAP need more than one training pathway? (affects *Pathway Structure Seam*)

Phase 2 — Accounts, Access & Email

Foundation · 7 features · **108h** · ~1.4 weeks · **Journey:** Step 1

Goal: The everyday account functions every real service needs: email, password reset, roles, offboarding.

Module	Feature	Hours
Email	Transactional Email Infrastructure	16
Auth	Password Reset	16
	Email Verification	14
	Session & Login Hardening	12

Module	Feature	Hours
Accounts	Account Settings	14
	Role Management Screen	20
	User Deactivation & Offboarding	16

Done when

- [ ] A locked-out user can reset their own password by email
- [ ] New accounts must verify their email address
- [ ] Staff can assign roles and deactivate users from the interface, with no developer involvement
- [ ] System email reliably reaches inboxes from the NFCAP domain

Phase 3 — Evidence & Competency ★

Core Product · 12 features · **260h** · ~3.3 weeks · **Journey:** Steps 8–10

Goal: Real photo evidence, assessed criterion by criterion. Certification only when every required criterion is Met.

Module	Feature	Hours
Evidence Storage	Real Evidence File Storage	32
	Image Downscaling & Storage Cost Control	12
Competency Model	Per-Course Assessment Criteria	18
	Criteria Authoring Interface	24
	Criteria Authoring Support (NFCAP)	12
Review	★ Reviewer Queue	24
	★ Per-Criterion Review & Gated Certification	40
	Photo Annotation	20
Learner	Guided Evidence Capture	28
	Per-Criterion Feedback & Targeted Resubmission	24
	Attempt History	12
QA	Legacy Data & Regression Pass	14

Done when

- [ ] A trainee uploads real photos against each named criterion and cannot submit until all required items are captured
- [ ] A reviewer assesses every criterion Met / Not Met with notes pinned on the photo
- [ ] ★ A certificate cannot be issued unless every required criterion is Met
- [ ] A rejected trainee sees exactly what failed and re-shoots only that

> ⚠️ **Depends on NFCAP:** trade experts must author the assessment criteria. The feature is only as credible as the criteria.

Phase 4 — Credential Integrity & Portability

Core Product · 4 features · **80h** · ~1.0 weeks · **Journey:** Step 10

Goal: A credential trainees can hold, prove and carry: a PDF certificate with public QR verification.

Module	Feature	Hours
Certificates	Downloadable PDF Certificate	20
	Public Verification Page + QR Code	20
	Expiry & Renewal Reminders	22
Portfolio	Shareable Verified Portfolio	18

Done when

- ☐ A trainee can download a PDF certificate
- ☐ Scanning a certificate's QR code publicly verifies it without a login
- ☐ Expiry and renewal behaviour is confirmed and, if required, implemented
- ☐ Portfolio sharing is either delivered or formally deferred by decision

> ⚠️ **Decision needed:** do credentials expire (renewal reminders)? Should trainees get a shareable public portfolio?

Phase 5 — Funding Module Productisation ★

Core Product · 7 features · **130h** · ~1.6 weeks · **Journey:** Steps 3–5

Goal: Make WIOA funding a clear, working process for trainees and staff: status, delivered email, impact reporting.

Module	Feature	Hours
Learner Experience	Funding Progress Transparency	28
	Eligibility Self-Check	20
Communications	Outreach Email Delivery & Deliverability	24
Case Management	Stalled-Case Detection & Follow-Up	20
Reporting	Funding Impact Reporting	22
Hardening	Document Upload Limits	8
	Funding Approval Sequencing Fix	8

Done when

- ☐ A trainee can see their funding stage, next step and who is responsible
- ☐ The self-check gives clear, correctly-qualified guidance before the full process
- ☐ Outreach email is genuinely delivered, with failures visible to staff
- ☐ Stalled cases are flagged and funding impact is reportable

Phase 6 — Communication & Operations

Operations & Access · 8 features · **184h** · ~2.3 weeks · **Journey:** Step 7, messaging

Goal: Day-to-day operations: messaging, announcements, cohorts, reporting and a quiz that cannot be gamed.

Module	Feature	Hours
Messaging	Trainee ↔ Staff Messaging	36
	Announcements	14
Notifications	Notification Coverage Completion	18
Operations	Bulk / CSV Enrollment	18

Module	Feature	Hours
	Cohorts & Deadlines	24
Reporting	Admin Reporting Depth	28
	Data Export	14
Assessment	Quiz Engine Depth	32

Done when

- [ ] A trainee can message staff from inside the system and get a reply
- [ ] Announcements reach trainees and key events reliably notify
- [ ] A cohort can be enrolled from a spreadsheet in one action
- [ ] Quiz retakes no longer reveal the answers, and reporting covers cohort trends

> ⚠️ **Decision needed:** how does NFCAP run cohort intakes? (affects *Cohorts & Deadlines*)

Phase 7 — Accessibility (WCAG 2.1 AA)

Operations & Access · 6 features · **102h** · ~1.3 weeks · **Journey:** All steps

Goal: Bring the whole interface to WCAG 2.1 AA, very likely required for WIOA-funded providers.

Module	Feature	Hours
Compliance	Confirm the Accessibility Obligation	8
Accessibility	Accessible Drag-and-Drop Reordering	20
	Dialog & Drawer Focus Handling	14
	Colour Contrast & Token Audit	16
	Semantics, Labels & Keyboard Pass	28
	Screen Reader Journey Testing	16

Done when

- [ ] The accessibility obligation is confirmed against primary sources
- [ ] Both primary journeys are fully operable by keyboard and screen reader
- [ ] Pathway reordering is accessible without a mouse
- [ ] Automated accessibility checks pass on every change

> ⚠️ **Confirm with NFCAP:** the exact accessibility obligation, checked against the regulation itself.

Phase 8 — Performance, Hardening & Launch

Launch · 6 features · **116h** · ~1.4 weeks · **Journey:** All steps (launch)

Goal: Prove it performs and is secure at scale, sign the data agreement, and go live.

Module	Feature	Hours
Performance	Load Testing & Database Tuning	24
	Caching & Media Delivery	16
Security	Independent Security Review	20
Operations	Runbook, Monitoring & Incident Process	16

Module	Feature	Hours
Compliance	Workforce Board Data Agreement (Gate)	12
Launch	User Acceptance Testing & Go-Live	28

Done when

- [ ] The system performs acceptably at NFCAP's expected scale
- [ ] An independent security review is passed and findings addressed
- [ ] The workforce board data agreement is signed and implemented
- [ ] Acceptance testing is signed off and the system is live with monitoring and a tested recovery path

> 🚧 **Hard gate:** the signed Workforce Board Data Agreement. Real WIOA data cannot enter the system without it.

7. Add-Ons (Quoted Separately)

Deliberately kept out of the base build — each gated on a client decision, or best scoped once the system is live. Multi-tenancy (supporting organisations beyond NFCAP) was **ruled out**: NFCAP confirmed single-organisation scope on 25 Oct 2026.

Add-on	Hours	Triggered when
Multi-Pathway / Multi-State Support	80	NFCAP needs more than one training pathway
Training Hour Tracking & DOL Reporting	120	The programme is DOL-registered
Self-Pay Checkout	60	Self-pay is a real enrolment route
Cross-Course Competency Framework	56	A skill proven once should count toward several courses
Advanced Analytics & At-Risk Prediction	48	After 6–12 months of live use
Mobile Evidence Capture (installable app, offline capture)	128	Trainees need to capture evidence on job sites with poor or no signal
External Credential Registry Integration	TBD	A registry is chosen
Native iOS / Android Apps	TBD	A true native app is wanted, after or instead of the installable option above
Subtotal (scoped items)	**492**	2 items TBD

8. Decisions Needed From NFCAP

Ordered by when each answer is needed, so no phase is blocked waiting.

Needed before	Topic	Decision
Now	Team size	Confirm the 2-engineer assumption behind the ~4-month estimate
Phase 1	Social Security proof	The required-document list includes proof of a Social Security number (card or W-2), which trainees would upload for staff review. The *No-SSN rule* says the system must not store one. Decide: accept the upload with extra protection, accept a different proof, or have the trainee show it to the board directly
Phase 1	Pathway count	Does NFCAP need more than one training pathway? *(also gates the Multi-Pathway add-on)*

Needed before	Topic	Decision
Phase 3	Assessment criteria	Trade experts available to author per-course criteria
Phase 4	Credential expiry	Do credentials expire and need renewal reminders?
Phase 4	Portfolio sharing	Should trainees get a shareable public portfolio? *(visible outside NFCAP if shared)*
Phase 5	Board checklists	Each workforce board has its own document checklist. Confirm which boards are needed at launch (only the Tri-Valley / Alameda County packet is on file) and that California is the only state
Phase 6	Intake model	How does NFCAP actually run cohort intakes?
Phase 7	Accessibility obligation	Confirm WCAG 2.1 AA against the regulation itself, not secondary sources
Phase 8	Scale target	Expected concurrent trainees and staff *(currently assumed: low thousands)*
Phase 8	Workforce Board agreement	Signed data-sharing and confidentiality agreement *(hard gate)*
Add-ons	DOL registration · self-pay route · credential registry · mobile app	See section 7